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ORIGINAL ARTICLE

Performance of a dosage individualization table for extended interval gentamicin in neonates beyond the first week of life

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Abstract

Objective: To assess the performance of a gentamicin dosing table for the individualization of extended-interval dosing (EID) in a neonatal population >7 days old.

Methods: A prospective observational study was carried out on gentamicin concentrations achieved using a dosing table in neonates >7 days old. Neonates were given 5 mg/kg IV gentamicin; then a table using 22 h post-first dose gentamicin concentrations was used to individualize dosing intervals. Pre- and post-serum gentamicin concentrations were measured and used to calculate the true peak and trough concentrations achieved.

Results: Use of the table resulted in dosing intervals that provided appropriate peak (mean 9.8 ± 1.8 mg/L) and trough (mean 0.6 ± 0.3 mg/L) concentrations in all neonates ($n = 38$). All trough concentrations were <2 mg/L, 83% were <1 mg/L. The majority of peak concentrations were in the usual target range (87%, 5–12 mg/L), with a few being in a higher, although likely safe range (13%, 12.1–15.7 mg/L).

Conclusions: Use of this dosing table to individualize extended-interval gentamicin dosages in neonates >7 days old resulted in appropriate peak and trough concentrations in all neonates studied. This allows appropriate extended-interval aminoglycoside dosages in neonates early in treatment.

Keywords

Aminoglycosides – administration and dosage, aminoglycosides – pharmacokinetics, Infant – newborn

History

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Introduction

Gentamicin, an aminoglycoside antibiotic, is one of the most frequently used antibiotics in the Neonatal Intensive Care Unit (NICU) for the treatment of neonatal sepsis [1].

There are two published meta-analyses on the use of extended-interval dosing (EID) of aminoglycosides in neonates [1,2]. Both analyses concluded that EID achieved target concentrations (defined in each study, most frequently trough ≤ 2 mg/L, peaks ≥ 5 mg/L) more often than traditional dosing. There were no differences in the incidence of nephrotoxicity or ototoxicity between the two regimens [1,2]. There was no difference with regards to treatment failures, although only a small number of neonates in the included studies had culture-proven sepsis [2].

While one of the advantages to EID is that measurement of serum gentamicin concentrations may be reduced or even unnecessary, given the highly variable nature of pharmacokinetics in neonates, many dosing regimens have had limited success in achieving target concentrations initially [3–7]. A mechanism for dosage individualization is needed, at least for

those patients who will receive aminoglycosides for longer than 48 h. Ideally this mechanism would require only one serum drug concentration. Several methods using a single serum concentration for dosage adjustment have been described [8–10].

When our regional NICUs switched to EID in 2009, selection of an initial dosing regimen was challenging due to the wide variety of dosages used in the literature [2–6,11–21]. Neofax[®] offered a simple method of dosage individualization, determining a dosing interval based on a 24 h post-first-dose concentration, however this method had not been previously validated in a clinical setting to our knowledge [10]. At our centre, it was thought that results from a 24 h concentration would be available too late to confirm a q24h dosing interval, so the Neofax[®] table was adjusted to use a 22-h post-dose concentration (Table 1). This table allowed us to determine if the patient required q24h, q36h or q48h dosing in time for the second dose, using a single 22 h level. Once use of the 22-h concentration dosing table was instituted in our NICUs, we completed a prospective validation of this dosing method in patients 7 days of age or younger with good results [22,23]. It was then necessary to confirm its performance in achieving target concentrations in infants over 7 days of age due to the changes that happen with advancing post-natal age (PNA) (i.e. decreasing total body water and improving renal function).

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Table 1. Suggested dosing interval based on 22-h concentration.

Level at 22 h (mg/L)	Interval (h)
≤1.2	24
1.3–2.6	36
2.7–3.5	48
≥3.6	Hold dose, repeat level in 24 h

Objective: The objective of our study was to evaluate the performance of a 22-h gentamicin concentration dosing table for the individualization of EID in neonates over 7 days of age, by examining the peak and trough concentrations achieved through its use.

Methods

Subjects

This study was performed in two Level II NICUs and one Level III NICU between September 2012 and December 2013. Approval by the local Research Ethics Review Board was received.

Infants over 7 days of age, who received at least two doses (5 mg/kg IV each) of EID gentamicin, were eligible for inclusion; infants of all gestational ages were included. The following serum gentamicin concentrations must have been available for inclusion: a level drawn 22-h after the start of the first dose, a pre-level drawn within 30 min prior to the start of the second or third doses and a post-level drawn 60–120 min after the start of the second or third doses (using the dosing interval indicated by the table). Patients considered to be unsuitable candidates for gentamicin (i.e. renal dysfunction, urine output <1 mL/kg/h) or hypoxic ischemic encephalopathy (HIE) are not given gentamicin in our NICUs, and so these patients were not eligible. Patients with concurrent use of indomethacin were excluded.

Gentamicin (at a concentration of 4 mg/mL) was given *via* infusion pump *via* an umbilical venous catheter (UVC), peripherally inserted central catheter (PICC) or peripheral intravenous line over 30 min. Normal saline flushes were performed before and after infusion. Bedside nurses drew blood for gentamicin concentration analysis *via* an arterial line, a capillary sample (heel stick) or a venous sample, always through a different line than the one used for gentamicin administration. Bedside nurses were also responsible for recording the timing of medication administration and blood draws; this was performed in the electronic patient care information system.

Data collection

Subjects were identified through the bedside physician or the clinical pharmacist on the team. Demographic, dosing and serum concentration information were collected through the patient care information system.

The neonatology team collected pre- and post-gentamicin concentrations in addition to the 22-h concentrations used with the table in infants over 7 days of age for approximately 2 years. The pre- and post-concentrations were measured with

the 2nd dose in infants on q36–48 h dosing, and with the 3rd dose in infants on q24h dosing. This timing was used to allow for any accumulation (i.e. approaching steady state) while ensuring that concentrations were done before the infant had completed their course of antibiotics. Serum drug concentrations were performed using a gentamicin assay on a Roche/Hitachi® Cobas C system (Rotkreuz, Switzerland), which has a functional sensitivity of 0.4 mg/L. Serum gentamicin assays were available 24 h a day, 7 days a week, with an approximate turnaround time of 1.5–2 h.

Data analysis

Based on the targets used in meta-analyses of EID in neonates, the achievement of trough concentrations less than 2 mg/L and peak concentrations greater than or equal to 5 mg/L using the 22-h concentrations dosing table can be considered successful [1,2]. An upper limit for peak concentrations was not considered in this study, although limits of 12 and 18 mg/L have been used in previous studies [11–13,14,18,19]. A target trough of <1 mg/L is also often suggested, so data for this target are presented as well [6–9,18,20,21].

Pharmacokinetic calculations were done using a linear, one-compartment model to calculate the volume of distribution (V_d), and half-life ($t_{1/2}$) in each patient. Peak concentrations were extrapolated to the end of the 30-min infusion for analysis. Mean and standard deviation were calculated for all data. Results were compiled for patients overall, by post-natal age, and by corrected gestational age (CGA).

Results

Data on a total of 48 courses of gentamicin were collected (46 patients). Data from 10 courses was excluded; six for missing data, two for errors in dosing or levels and two for patients not meeting age criteria. A total of 38 courses were analyzed. The median length of the included gentamicin courses was 6 days (range 1–15 days), 82% of courses were a week or less in duration.

Gestational ages at birth ranged from 24 weeks to 36 weeks and 2 days; the median was 27.1 (IQR 25.4–28.7) weeks. Post-natal ages at the start of gentamicin ranged from 8 to 95 days; the median was 26 (IQR 12–40) days (30%, 8–14 d; 23%, 14–28 d; 35%, 28–60 d; 12%, >60 d). The corrected gestational age at start of gentamicin ranged from 25 weeks and 3 days to 40 weeks and 6 days; the median was 31.9 (IQR 29.3–34.6) weeks (20%, ≤28 wks; 50%, 29–32 wks; 12%, 33–36 wks; 18%, ≥37 wks).

Using dosing intervals suggested by the 22-h concentration dosing table, all patients had extrapolated trough (i.e. C_{min}) concentrations <2 mg/L (range <0.4–1.3 mg/L) before the 2nd or 3rd dose; 83% of troughs were <1 mg/L (Figure 1). Because 30% trough levels were undetectable (i.e. <0.4 mg/L) we were unable to calculate the true mean trough. Substituting in the maximum (0.3 mg/L) and minimum (0 mg/L) values possible for these troughs results in a mean trough range of $0.5–0.6 \pm 0.4$ mg/L.

When calculating extrapolated peaks (C_{max}), we used 0.3 mg/L as the value for troughs that were undetectable; this may underestimate true peak values. Using this method all

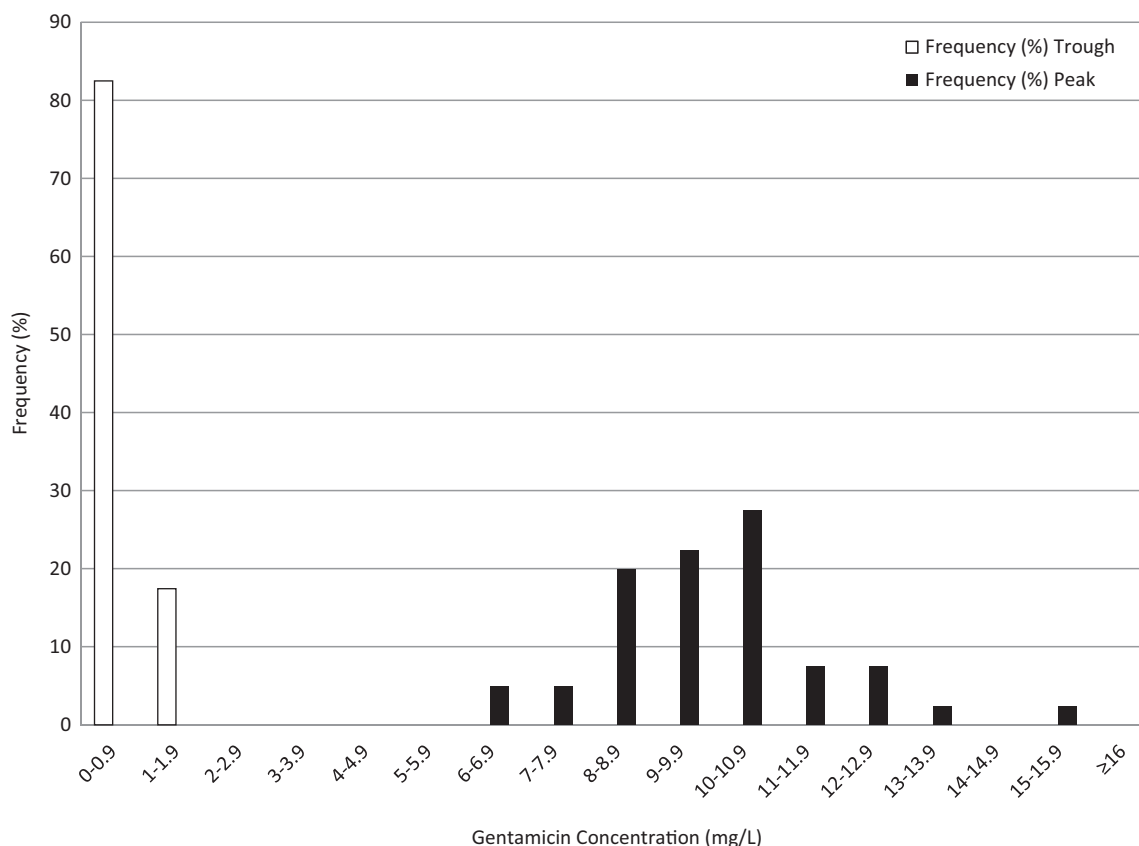


Figure 1. Frequency of trough and peak gentamicin levels achieved using 22 h level dosing table.

Table 2. Results by gestational age and post-natal age at start of gentamicin.

		Peaks (mg/L) Mean (SD)	Troughs (mg/L) Mean (SD) (using 0.3 for undetectable levels)	Vd (L/kg) Mean (SD)	Half-life (h) Mean (SD)
By CGA	≤28 wks (n = 8)	10.4 (2.6)	0.9 (0.3)	0.55 (0.17)	10.2 (3.5)
	29–32 wks (n = 20)	9.7 (1.7)	0.6 (0.3)	0.56 (0.13)	6.5 (1.6)
	33–36 wks (n = 5)	10.0 (1.9)	0.7 (0.4)	0.55 (0.08)	5.9 (1.4)
	≥37 wks (n = 7)	9.6 (0.8)	0.4 (0.2)	0.55 (0.09)	5.0 (0.5)
By PNA	7–14 d (n = 12)	10.3 (1.8)	0.8 (0.3)	0.54 (0.14)	8.8 (3.0)
	14–28 d (n = 9)	9.2 (1.0)	0.7 (0.3)	0.61 (0.10)	7.2 (3.2)
	28–60 d (n = 14)	10.2 (1.6)	0.6 (0.3)	0.52 (0.11)	5.8 (1.3)
	>60 d (n = 5)	8.9 (0.9)	0.4 (0.2)	0.61 (0.08)	5.1 (0.6)
Overall (n = 40)		9.8 (1.8)	0.6 (0.3)	0.56 (0.12)	6.9 (2.6)

patients had peaks >5 mg/L after the 2nd or 3rd dose; 87% had a peak in the 5–12 mg/L range, and the remaining 13% had concentrations ranging from 12.1 to 15.7 mg/L (Figure 1). The mean peak concentration was 9.8 ± 1.8 mg/L, the range was 6.2–15.7 mg/L. Variability in peaks and troughs based on PNA and CGA was minimal (Table 2).

The mean calculated Vd for the study group was 0.56 ± 0.12 L/kg; the range was 0.2–0.9 L/kg. Mean $t_{1/2}$ was 6.9 ± 2.6 h, ranging from 4.4 to 15.3 h. Again, these values were calculated using 0.3 mg/L as the value for undetectable trough values. This may overestimate $t_{1/2}$ and Vd. Vd stayed relatively constant over CGA and PNA; $t_{1/2}$ decreased with increasing PNA and CGA (Table 2).

Nephrotoxicity monitoring in our population consists of diligent urine output measurement daily, with regular serum creatinine (SCr) monitoring (frequency depending on gestational age and clinical condition). All patients had normal urine output (>1 mL/kg/h) during gentamicin therapy and throughout the remainder of their admission. Two patients had an elevated (>0.68 mg/dL) SCr measurement while on gentamicin. One patient had a SCr value of 0.94 mg/dL on day 2 of gentamicin therapy (SCr 11 days prior was 0.59 mg/dL); a repeat SCr one month later had returned to normal (0.39 mg/dL). Another patient had a SCr of 1.52 mg/dL on day 5 of gentamicin. This patient had a previously elevated SCr of 0.97 mg/dL 5 days prior to starting

gentamicin, and had repeated elevations in SCr while in the NICU not temporally related to gentamicin. SCr was normal at discharge.

With regards to ototoxicity, 30 patients underwent hearing assessments prior to discharge; one infant did not pass. Hearing loss in the NICU is a multi-factorial issue and as such we were unable to assess if our dosing scheme contributed to this patient's hearing loss.

Discussion

We have demonstrated that use of a 5 mg/kg IV gentamicin dose individualized with a dosing interval determined by the 22-h concentration dosing table resulted in appropriate peak and trough concentrations (with the 2nd or 3rd dose) in neonates older than 7 days of age, as previously shown in infants 7 days of age or less [22,23]. This confirms that the dosing table accounted for the variable renal function ($t_{1/2}$) and Vd present in the wide range of gestational and post-natal ages.

We used a one-compartment model to extrapolate gentamicin trough ($C_{\min}$) and peak ($C_{\max}$) values from pre- and post-concentrations drawn by NICU staff. Mean measured pre- and post-concentrations were very similar to the extrapolated trough/ $C_{\min}$ and peak/ $C_{\max}$.

Aminoglycosides display concentration-dependent killing, suggesting higher peaks provide greater efficacy [7,24,25]. While a peak aminoglycoside concentration to minimum inhibitory concentration (MIC) ratio of 8–10:1 is considered ideal, based on the usual MICs of *Escherichia coli* (range 0.25–1 mg/L) a peak of at least 5 mg/L has a high likelihood of being effective [2,25]. A maximum peak concentration of 12 mg/L is frequently cited in the literature [1,7], although a maximum of up to 18 mg/L has been used [13]. Our patients' peak levels were within these limits 88% and 100% of the time, respectively. Our calculated peak values would be higher if we used a value of zero for the undetectable levels, however higher peaks are not considered more likely to cause toxicity due to saturable uptake of aminoglycosides in the renal cortex and perilymph [2,7,24,26–29]. Indeed, adult patients on EID of aminoglycosides achieve peaks of 20 mg/L, with evidence of less nephrotoxicity as compared to traditionally dosed gentamicin [7,26–29]. EID reduces accumulation of gentamicin in the renal cortex by 30–50%, as compared to traditional dosing, reducing the potential for nephrotoxicity [7,24,26–29].

Nephrotoxicity of gentamicin is more closely associated with trough concentrations greater than 2 mg/L [28,29]. All neonates in our study had trough concentrations below this limit. A target concentration of <1 mg/L for EID gentamicin is also often cited as a target trough; the majority of our patients (81%) achieved this target as well [29]. Only two of the 38 patients included had elevated serum creatinine while on gentamicin, and only one of these appears to be temporally related to gentamicin. Estimation of area under the concentration-time curve (AUC) was beyond the scope of our study; however some authors have related area under the curve values to aminoglycoside nephrotoxicity [30]. EID modulates the effect of AUC on the incidence of nephrotoxicity in that the probability of nephrotoxicity at the same AUC value is less

with EID than it is with traditional aminoglycoside dosing [30]. Our units' previous traditional gentamicin dosing used doses of 2.5 mg/kg IV q12–24 h, which provided total daily doses equivalent to our EID regimen (2.5–5 mg/kg/day), thus we would not expect significantly different AUC values with this new regimen. Our unit's approach to minimizing aminoglycoside toxicity is first to avoid giving gentamicin to patients at elevated risk (i.e. on indomethacin, history of hypoxia and/or significant renal dysfunction), secondly to minimize the duration of treatment and finally to dose gentamicin in a way that minimizes risk (i.e. EID with dose adjustment as necessary) [31].

Aminoglycosides display a post-antibiotic effect whereby bacterial growth is suppressed despite negligible drug concentrations [27]. Low glomerular filtration rates in premature neonates may dictate prolonged dosing intervals (q36–48 h), thus creating the potential for prolonged periods with negligible gentamicin concentrations [18,24,25,32]. In this cohort of infants 7 days of age or older, the majority of infants (78%) were placed on a 24 h dosing interval, which would minimize the duration of the drug-free interval and any potential bacterial re-growth at the end of longer dosing intervals [27]. Higher peak concentrations are thought to prolong the post-antibiotic effect, and meta-analyses on EID of aminoglycosides in neonates have not shown reduced efficacy as compared to traditional dosing [1,2].

When a mechanism for individualizing gentamicin dosages is necessary (for example in neonates expected to be on gentamicin for longer than 48–72 h), the method should ideally require a single serum drug concentration. Some NICUs prefer to measure trough concentrations, however trough concentration results are often available only after the next dose has been given, and guidance on how to adjust dosages using trough concentrations is limited in the literature. Others have described the use of nomograms to adjust EID gentamicin in neonates, however to our knowledge these nomograms have not been prospectively validated to date [8,9].

A recent study using a dosing schedule similar to ours (4 mg/kg IV given q24–48 h based on gestational age) showed success in achieving target concentrations [21]. After the first dose, the dosing scheme resulted in peaks ≥ 6 mg/L in 87% patients, and troughs <2 mg/L in 98% patients [21]. When concentrations were repeated with the fourth infusion, troughs remained <2 mg/L (99%), but the percentage of patients with peaks ≥ 6 mg/L decreased to 55% [21]. This decrease in peak concentrations was associated with fluid retention and marked increases in body weight [21]. While our study did not include repeated concentrations with the fourth dose of gentamicin, these results highlight the need to recheck serum drug concentrations with prolonged courses of gentamicin, or when significant fluid shifts or changes in renal function occur.

While the advantage of nomograms is the flexibility in the timing of serum drug concentrations, use of a specifically-timed 22 h serum concentration in our units has not been a barrier to use of this dosing system. Our nursing units' practices have adapted such that other necessary blood work (i.e. C-reactive protein) is performed with the 22 h gentamicin concentrations in order to reduce excessive blood draws. The

nursing and physician teams find this dosing system practical in that it allows for the determination of dosing intervals before the 2nd dose is due, potentially avoiding excessive dosing early on in the course of treatment.

Because performance of the dosing table has only been measured with use early on in treatment (i.e. with the 2nd or 3rd dose) its use may be most appropriate for short courses (i.e. <7 days) of gentamicin. This reflects the vast majority of gentamicin used in our NICUs. It is also our practice to draw pre- (trough) gentamicin concentrations on a weekly basis when courses are prolonged (i.e. >10 d) to ensure accumulation has not occurred. This may help to ensure appropriate concentrations for maintaining efficacy and avoiding toxicity in this population. Our study was also limited by its reliance on typical renal monitoring practices, which may not be sufficient for patients on prolonged courses of gentamicin.

Ideally, clinical validation of this dosing method would be performed in a larger number of infants of varying corrected-gestational and post-natal ages. However, our unit rarely uses gentamicin in patients over 7 days of age and so obtaining a larger numbers of patients was not feasible. The data we do have are reassuring, however, given that most patients achieved target levels, and all achieved levels likely to be safe and effective. Diligent monitoring of serum creatinine, urine output, cultures and signs of infection in addition to the use of the dosing interval adjustment table will maximize gentamicin's clinical efficacy and safety.

Conclusion

This study evaluated the use of a gentamicin dosage individualization table, specifically using 22 h post-first dose serum gentamicin concentrations to determine an appropriate dosing interval for EID in neonates over 7 days of age. This is important as $t_{1/2}$ and V_d changes occur rapidly throughout the neonatal period. The results showed that the dosing table provided dosing intervals that achieved appropriate peak (mean 9.8 mg/L) and trough (mean 0.6 mg/L) concentrations in most subjects. All subjects had trough concentrations <2 mg/L, 83% were <1 mg/L; all peaks were >5 mg/L. Because assessment of the table's performance was done using concentrations measured early in treatment (i.e. with the 2nd or 3rd doses), its use is most appropriate for short courses of gentamicin.

This method provides a practical way of individualizing EID aminoglycoside dosages in neonates (including very low birth weight infants) beyond the first week of life. Advantages to the use of the table include the ability to determine appropriate dosing intervals based on one concentration only, early in treatment.

Declaration of interest

The authors report no declarations of interest.

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